

Circuit Documentation

1. Circuit Summary

This circuit is an Arduino Mega 2560–controlled mobile/robotic system consisting of:

- A 3-cell, 2200 mAh battery pack as the primary power source.
- An XL4016 adjustable DC-DC converter.
- An L298N dual H-bridge motor driver.
- Two 37 mm DC motors controlled by the L298N.
- Six MG996R servo motors powered from the XL4016 output.
- An EBYTE E01-ML01DP5 wireless module connected through an NRF24L01 5 V adapter.
- A rocker switch used to connect and disconnect the battery return path.
- A two-pin terminal block providing an additional connection point on the servo power rails.
- An Arduino Mega 2560 providing motor-driver, servo, and wireless-module control signals.

The two DC motors are connected to the two channels of the L298N motor driver. The Arduino controls the motor-driver enable and direction inputs. The six MG996R servos receive power from the XL4016 output and are controlled by six Arduino digital pins.

The wireless module and its adapter share the SPI bus and control signals. The adapter's CE and CSN signals are connected to Arduino Mega pins D2 and D3, while the hardware SPI signals use pins D50–D52.

The supplied Arduino code is only the default empty sketch. It does not currently initialize the hardware, control the motors or servos, or communicate with the wireless module.

Important electrical observations

1. The XL4016 output voltage is not specified in the provided design. It must be adjusted and verified before connecting the Arduino, wireless adapter, or servos.
2. MG996R servos can draw substantial stall and startup current. The XL4016 and wiring must be sized for the combined servo load.
3. The L298N 5V pin is not listed as connected. The required logic supply arrangement for the specific L298N module should be verified.
4. The NRF24L01 adapter VCC is connected to the same XL4016 output as the servos and Arduino VIN. Confirm that this adapter accepts the selected voltage.
5. The battery negative terminal is switched using the rocker switch. This is a low-side power switch rather than the more conventional positive-side arrangement.

2. Component List

Arduino Mega 2560

The Arduino Mega 2560 is the main controller. It provides:

- Six servo control outputs.
- Six motor-driver control outputs.
- SPI communication with the wireless module.

- Shared power and ground connections with the rest of the circuit.

L298N Dual H-Bridge Motor Driver

The L298N controls two brushed DC motors. It provides:

- Direction control for each motor.
- Enable/PWM control for each motor channel.
- Separate output pairs for the two motors.
- Motor supply input through +12 and GND.

XL4016 DC-DC Converter

The XL4016 regulates the battery voltage and supplies the downstream electronics and servos through its OUT+ and OUT- terminals.

Its output voltage is not specified and must be configured according to the requirements of all connected loads.

3S 2200 mAh Battery

The 3S battery is the primary energy source. A 3S lithium battery typically has:

- Approximately 11.1 V nominal voltage.
- Approximately 12.6 V when fully charged.

The battery positive terminal feeds the XL4016 input and the L298N motor supply. The battery negative terminal is routed through the rocker switch.

Rocker Switch

The rocker switch disconnects the battery negative return from the system ground when turned off.

37 mm DC Motor (Motor 1)

The first DC motor is driven by L298N channel A through OUT-1 and OUT-2.

37 mm DC Motor (Motor 2)

The second DC motor is driven by L298N channel B through OUT-3 and OUT-4.

MG996R Servo Motors

Six MG996R servos are connected in parallel to the XL4016 output power rails. Each servo has:

- VCC
- GND
- Signal

The servo signal wires are individually connected to Arduino pins D6 through D12, excluding D13.

NRF24L01 5V Adapter

The adapter provides an interface and power connection for the wireless module. Its signal pins are connected to the Arduino Mega SPI bus and two general-purpose digital pins.

EBYTE E01-ML01DP5 Wireless Module

The EBYTE module is connected to the NRF24L01 adapter pin-for-pin for:

- Power
- Ground
- CE
- CSN
- SPI
- IRQ

Terminal PCB 2 Pin

The two-pin terminal is connected across the servo supply:

- Pin A: XL4016 output negative
- Pin B: XL4016 output positive

It provides an external access point for the servo power rails.

Comment V2 Objects

The Comment V2 objects contain no electrical pins and do not form part of the electrical circuit. They are documentation or annotation elements only.

3. Component Wiring Details

3.1 Arduino Mega 2560

Power

- VIN → XL4016 OUT+
- GND → XL4016 OUT-

Wireless-module control and SPI

- D2 → NRF24L01 adapter CE
- D3 → NRF24L01 adapter CSN
- D50 → NRF24L01 adapter MISO
- D51 → NRF24L01 adapter MOSI
- D52 → NRF24L01 adapter SCK

Motor-driver control

- D4 → L298N EN-A
- D5 → L298N EN-B
- D22 → L298N IN-1
- D24 → L298N IN-2
- D26 → L298N IN-3
- D28 → L298N IN-4

Servo control

- D6 → MG996R signal, unnumbered MG996R
- D7 → MG996R (1) Signal
- D8 → MG996R (5) Signal
- D9 → MG996R (6) Signal
- D10 → MG996R (4) Signal
- D11 → MG996R (3) Signal
- D12 → MG996R (2) Signal

The design contains seven servo signal connections in the net list, although the parts list contains six MG996R servo instances. Specifically, the D6 connection is associated with an additional unnumbered MG996R, while D7 through D12 connect to six numbered or otherwise identified servo instances. This discrepancy should be resolved before fabrication.

3.2 L298N Dual H-Bridge Motor Driver

Power

- +12 → Battery positive and XL4016 IN+
- GND → XL4016 IN-, rocker switch pin 1, and system ground
- 5V → No connection listed

Control inputs

- EN-A → Arduino Mega D4
- IN-1 → Arduino Mega D22
- IN-2 → Arduino Mega D24
- IN-3 → Arduino Mega D26
- IN-4 → Arduino Mega D28
- EN-B → Arduino Mega D5

Motor outputs

- OUT-1 → Motor 1 +
- OUT-2 → Motor 1 -
- OUT-3 → Motor 2 +

- OUT-4 → Motor 2 -

The motor polarity is defined by the connection list. Reversing the two wires for a motor will reverse the relationship between the Arduino direction commands and the physical motor rotation.

3.3 XL4016 DC-DC Converter

Input

- IN+ → Battery + and L298N +12
- IN- → Rocker switch pin 1 and L298N GND

Output

- OUT+ → Arduino Mega VIN
- OUT+ → All six MG996R VCC pins
- OUT+ → NRF24L01 adapter VCC
- OUT+ → Terminal PCB Pin B
- OUT- → Arduino Mega GND
- OUT- → All six MG996R GND pins
- OUT- → NRF24L01 adapter GND
- OUT- → Terminal PCB Pin A

The converter output voltage must be measured with a multimeter before connecting the load. The correct voltage depends on the requirements of the Arduino input, servo motors, and wireless adapter.

3.4 3S 2200 mAh Battery

- + → XL4016 IN+
- + → L298N +12
- - → Rocker switch pin 2

The battery negative terminal does not directly connect to system ground; it reaches system ground through the rocker switch.

3.5 Rocker Switch

- Pin 2 → Battery -
- Pin 1 → XL4016 IN- and L298N GND

Because the switch is placed in the negative return path, it switches the system ground connection. For improved safety and more conventional power isolation, a positive-side switch or a properly rated double-pole switch may be preferable.

3.6 37 mm DC Motor (Motor 1)

- + → L298N OUT-1
- - → L298N OUT-2

This motor is controlled by L298N channel A.

3.7 37 mm DC Motor (Motor 2)

- + → L298N OUT-3
- - → L298N OUT-4

This motor is controlled by L298N channel B.

3.8 MG996R Servo Motors

All six servos share the same supply rails.

Common servo power connections

- Each servo VCC → XL4016 OUT+
- Each servo GND → XL4016 OUT-

Individual signal connections

Servo	Signal connection
MG996R connected to D6	Arduino Mega D6
MG996R (1)	Arduino Mega D7
MG996R (5)	Arduino Mega D8
MG996R (6)	Arduino Mega D9
MG996R (4)	Arduino Mega D10
MG996R (3)	Arduino Mega D11
MG996R (2)	Arduino Mega D12

The parts list identifies six MG996R components, but the net list identifies seven servo signal connections. The extra D6 servo connection should be checked against the intended hardware inventory.

3.9 NRF24L01 5V Adapter

Power

- VCC → XL4016 OUT+
- GND → XL4016 OUT-

Control and SPI signals

- CE → EBYTE module CE and Arduino Mega D2
- CSN → EBYTE module CSN and Arduino Mega D3
- SCK → EBYTE module SCK and Arduino Mega D52
- MOSI → EBYTE module MOSI and Arduino Mega D51
- MISO → EBYTE module MISO and Arduino Mega D50
- IRQ → EBYTE module IRQ

The adapter IRQ pin is connected only to the EBYTE module in the provided net list. It is not connected to an Arduino pin.

3.10 EBYTE E01-ML01DP5 Wireless Module

The EBYTE module is connected to the NRF24L01 adapter as follows:

- VCC ↔ NRF24L01 adapter VCC
- GND ↔ NRF24L01 adapter GND
- CE ↔ NRF24L01 adapter CE
- CSN ↔ NRF24L01 adapter CSN
- SCK ↔ NRF24L01 adapter SCK
- MOSI ↔ NRF24L01 adapter MOSI
- MISO ↔ NRF24L01 adapter MISO
- IRQ ↔ NRF24L01 adapter IRQ

The Arduino communicates with the adapter, which passes the relevant signals to the EBYTE module.

3.11 Terminal PCB 2 Pin

- Pin A → XL4016 OUT-
- Pin B → XL4016 OUT+

This terminal provides an auxiliary connection to the regulated output supply.

4. Documented Code

4.1 Supplied Firmware

The supplied firmware is an empty Arduino sketch:

```
void setup() {  
  // put your setup code here, to run once:
```

```

}

void loop() {
  // put your main code here, to run repeatedly:

}

```

4.2 Current Firmware Functionality

The current code does not perform any hardware functions:

- No pin modes are configured.
- No motor-driver outputs are initialized.
- No PWM or enable signals are generated.
- No servo objects are created or attached.
- No servo positions are commanded.
- No SPI interface is initialized.
- No wireless-library configuration is present.
- No wireless data is transmitted or received.
- No safety or startup behavior is implemented.

All Arduino pins therefore remain in their default reset state until additional firmware is added.

4.3 Intended Pin Assignment for Future Firmware

The following assignments can be used when implementing the control software:

```

// Wireless module
#define RADIO_CE    2
#define RADIO_CSN   3
// Hardware SPI on Mega:
// MISO = 50
// MOSI = 51
// SCK  = 52

// L298N motor driver
#define MOTOR_A_EN  4
#define MOTOR_B_EN  5
#define MOTOR_IN1   22
#define MOTOR_IN2   24
#define MOTOR_IN3   26
#define MOTOR_IN4   28

// Servo outputs
#define SERVO_1_PIN  6
#define SERVO_2_PIN  7
#define SERVO_3_PIN  8

```

```
#define SERVO_4_PIN 9
#define SERVO_5_PIN 10
#define SERVO_6_PIN 11
#define SERVO_7_PIN 12
```

The seven listed servo outputs should be reconciled with the six MG996R parts listed in the circuit design.

4.4 Recommended Firmware Startup Behavior

Before operating the circuit, firmware should:

1. Configure all L298N input and enable pins as outputs.
2. Set both motor enable pins low during startup.
3. Set all motor direction inputs to a known stopped state.
4. Attach and initialize the servo outputs.
5. Set servos to safe initial positions.
6. Initialize the wireless module using the selected CE and CSN pins.
7. Verify wireless communication before accepting motion commands.
8. Include a timeout that stops the motors if valid control packets are not received.
9. Avoid commanding motors or servos until power and communication initialization has completed.